

ECO Unit Queue Management System

Staff Web Portal

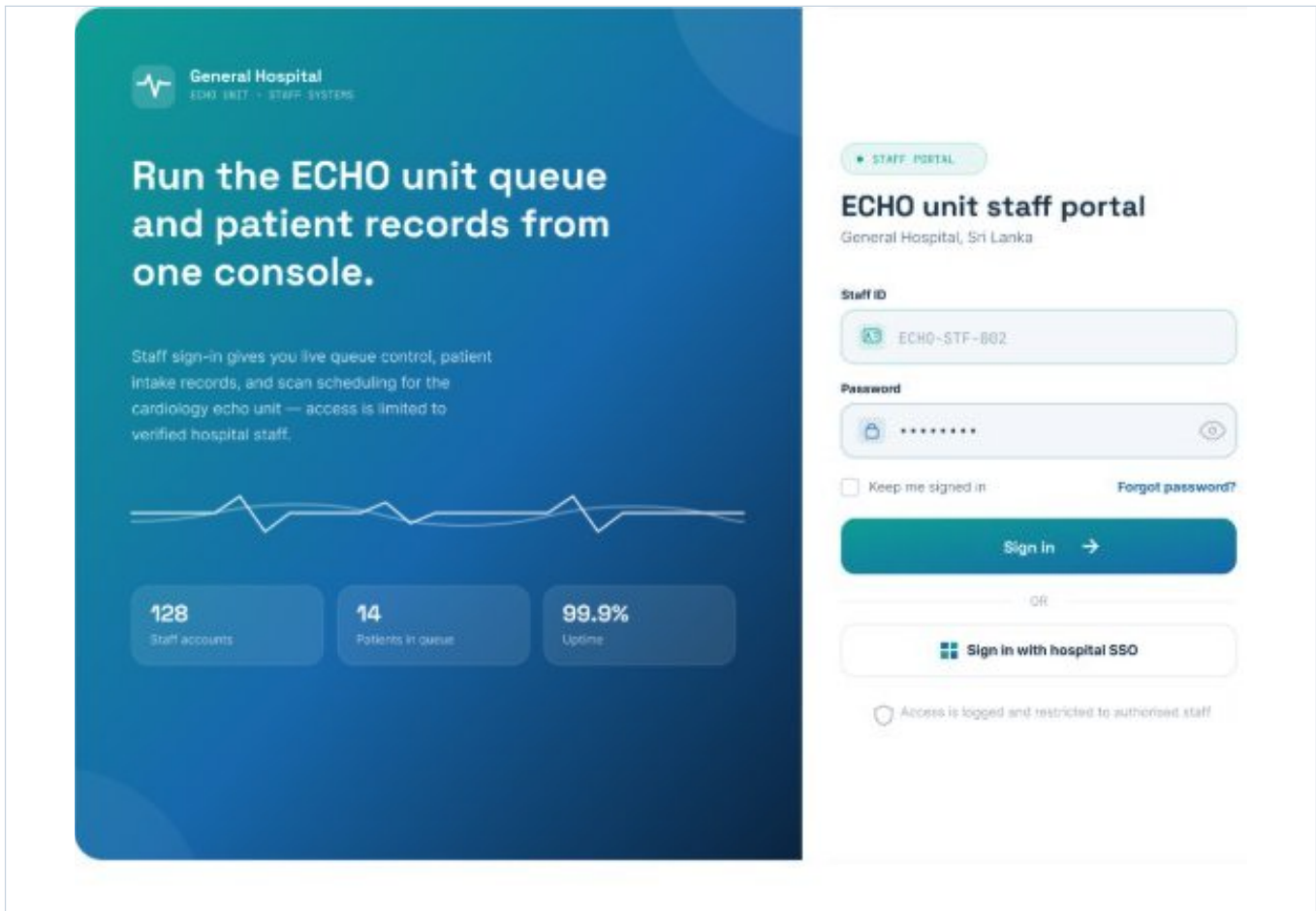
UI Screens — Actor, Purpose, Functions and Requirement Mapping

This document catalogues every implemented screen of the Staff Web Portal for the ECO (Echocardiography) unit queue-management system described in the accompanying research proposal (General Hospital, Sri Lanka). For each screen it identifies the primary actor (user role), the screen's purpose, its key functional elements, and how it maps to the functional requirements (FR1–FR12, Table 3.1) and operational problems identified in Chapter 1 of the proposal.

Primary actor for this portal: Staff — front-desk/registration clerks, nursing staff, and the medical officers/technicians performing echocardiograms in the ECO unit.

Screens documented: 11 (Staff Sign In, Forgot/Reset Password, Today's Overview, Queue Management, Queue Prediction, Patient Search, Patient Detail, Reports, Register New Patient modal, My Profile, Edit Profile).

Screen 1. Staff Sign In



Actor: Staff (front-desk/registration staff, nurses, ECHO unit technicians, and medical officers) — any authorised ECO unit staff member

Purpose

This is the entry point of the Staff Web Portal. It authenticates hospital staff before granting access to any patient data, queue controls, or reporting tools, so that the system is used only by verified ECO unit personnel.

Key functions / elements

- Staff ID and password fields, with a show/hide toggle on the password field
- “Keep me signed in” checkbox for trusted devices
- “Forgot password?” link that opens the password-reset flow (Screen 2)
- Primary “Sign in” action button plus an alternative “Sign in with hospital SSO” option
- A visible security notice (“Access is logged and restricted to authorised staff”) reinforcing accountability
- A branded panel summarising what the portal offers (live queue control, patient intake records, scan scheduling) together with live-style indicator cards (staff accounts, patients in queue, uptime)

Relation to requirements / research proposal

Supports the non-functional security and data-privacy requirements described in the methodology (Section 3.10): access to patient data must be restricted and every sign-in/reset event is logged for audit purposes, feeding into the Audit Log maintained on the Administrator Web Portal. It is the gatekeeper for every functional requirement (FR1–FR12) that follows, since none of the staff-side features can be reached without authentication.

Screen 2. Forgot Password / Reset Password

General Hospital
ECHO UNIT - STAFF SYSTEMS

Lost access to your staff account? We'll get you back in safely.

Password resets are sent only to the verified email on file for your Staff ID. Reset links expire after 15 minutes for the security of patient records on the cardiology echo unit.

128 Staff accounts 15 min Link expiry 99.9% Uptime

Reset your password
General Hospital, Sri Lanka

Enter your Staff ID below. If it matches an active account, we'll email a secure reset link to the address on file.

Staff ID

ECHO-STF-082

Reset links are single-use and expire after 15 minutes.

Send reset link

[Back to sign in](#)

Password resets are logged for audit and security review

Actor: Staff (any registered user who has lost access to their account)

Purpose

Provides a self-service account-recovery path so that a staff member who cannot sign in is not permanently locked out, while keeping the recovery process secure and auditable.

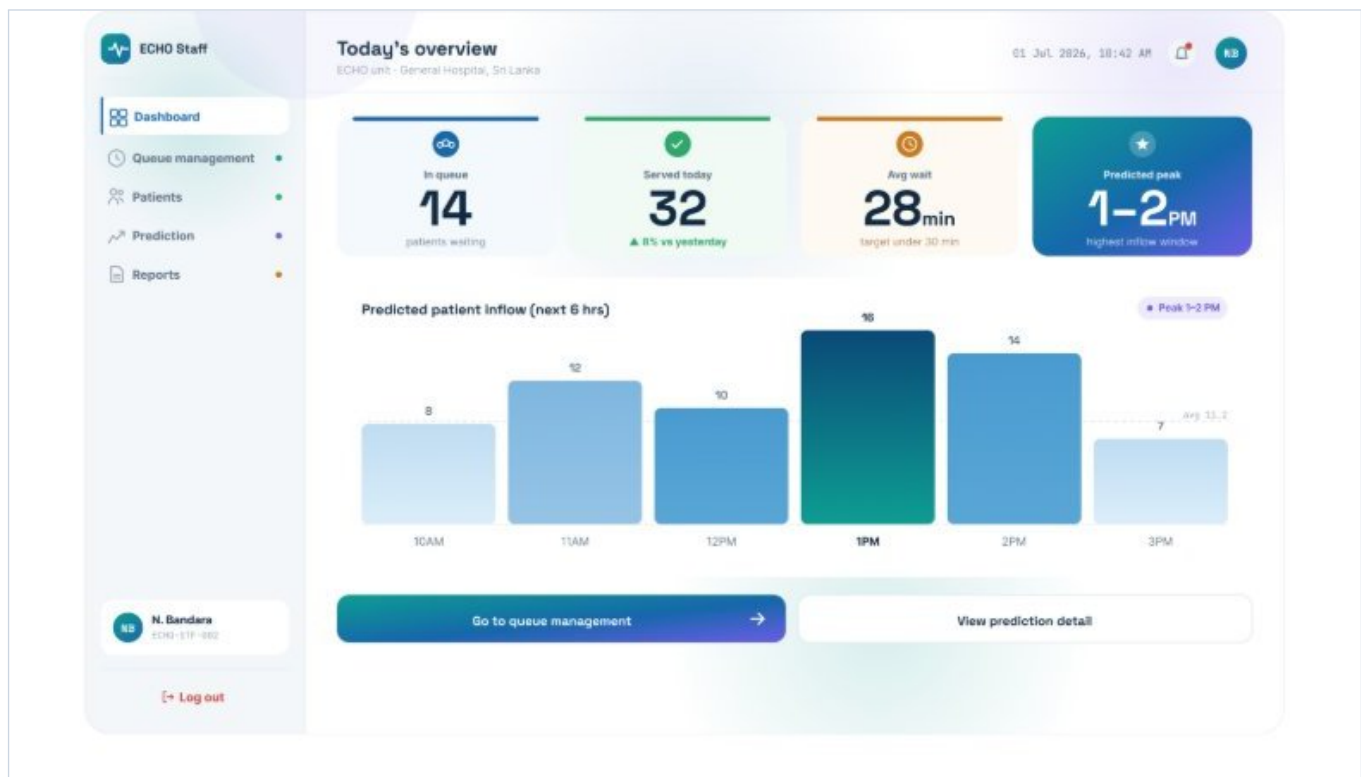
Key functions / elements

- Explanatory panel describing that resets are only ever sent to the verified email on file for the entered Staff ID
- Staff ID input field to identify the account requiring recovery
- Inline notice that the reset link is single-use and expires after 15 minutes
- “Send reset link” primary action and a “Back to sign in” link to return to Screen 1
- A footer notice confirming that password resets are logged for audit and security review

Relation to requirements / research proposal

Directly implements the security and account-governance safeguards discussed in Section 3.10 (Ethical Considerations) of the proposal, where role-based access, session control, and audit logging are identified as required controls for any system handling patient-adjacent data under Sri Lanka’s Personal Data Protection Act, No. 9 of 2022.

Screen 3. Today's Overview (Dashboard)



Actor: Staff (front-desk/registration staff and ECHO unit technicians) — landing screen after login

Purpose

Gives staff an at-a-glance operational snapshot of the ECO unit the moment they log in, so they can start their shift with situational awareness instead of walking in blind, directly answering the “lack of real-time monitoring” problem identified in Chapter 1.

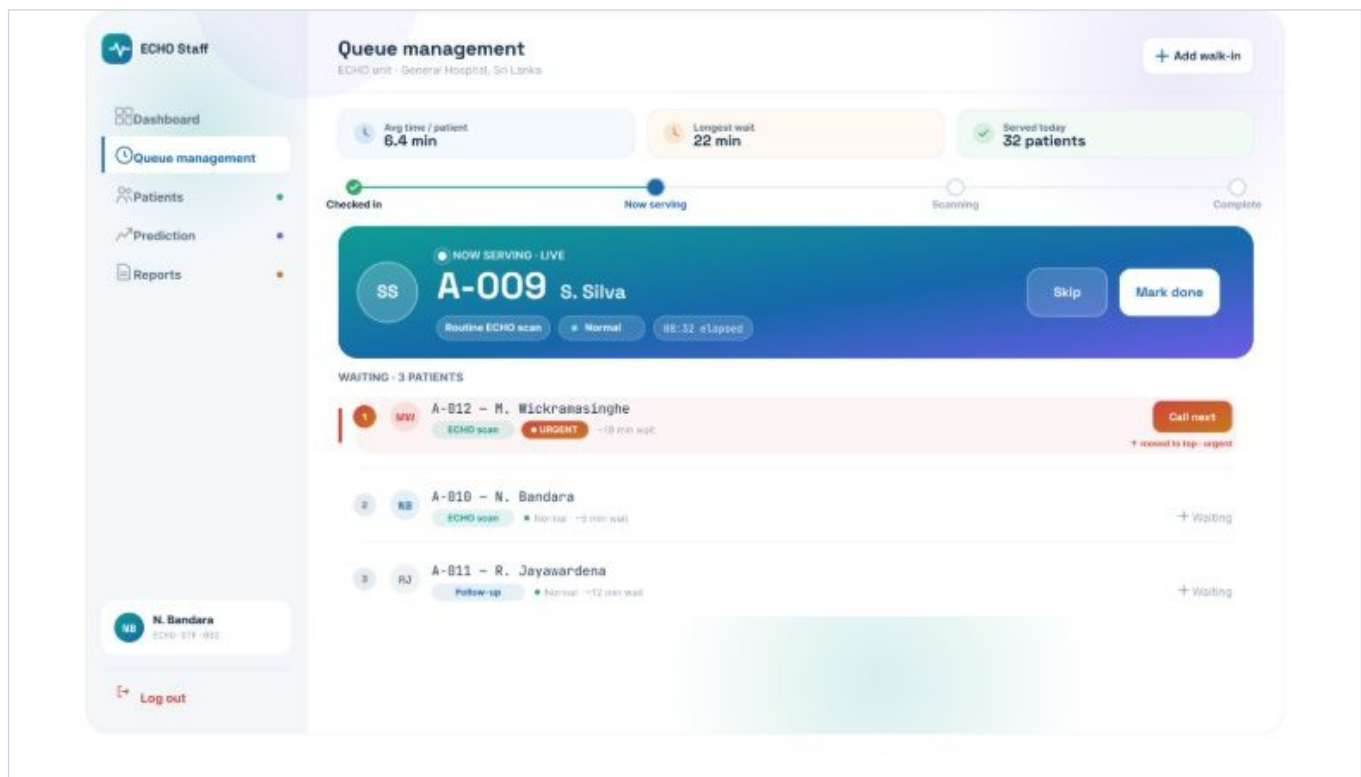
Key functions / elements

- Four key-performance-indicator cards: patients currently In queue, patients Served today (with day-on-day % change), Avg wait (against a target), and the ML-predicted Peak window for the day
- “Predicted patient inflow (next 6 hrs)” bar chart showing hourly expected patient volume, with the current/peak hour highlighted
- Quick-action buttons: “Go to queue management” and “View prediction detail”, linking directly into Screens 4 and 5
- Persistent left-hand navigation (Dashboard, Queue management, Patients, Prediction, Reports) and a header with date/time, notifications, and the signed-in staff avatar

Relation to requirements / research proposal

Corresponds to the “Today’s Overview dashboard with key-performance-indicator cards and a patient-volume chart” listed as a core Staff Web Portal screen in the Scope of the Study (Section 1.5), and is fed by the same ML Prediction Service (Figure 3.2) that powers the Queue Prediction screen.

Screen 4. Queue Management



Actor: Staff (front-desk staff and ECHO unit technicians operating the live queue)

Purpose

The core operational control surface of the portal: it lets staff run the live ECO queue in real time — calling patients, skipping them, marking scans complete, and pulling urgent cases forward — replacing the informal, memory-based ordering used in the manual process.

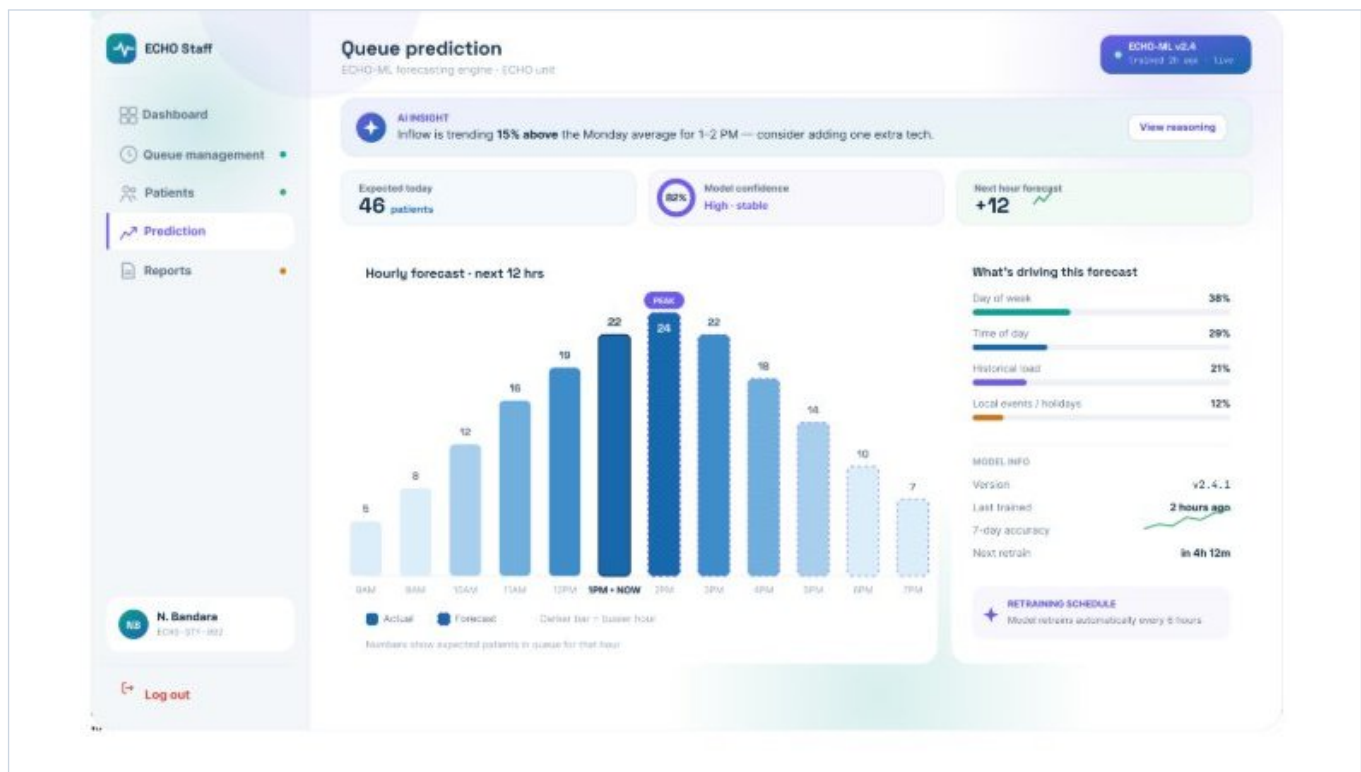
Key functions / elements

- Summary strip: Avg time/patient, Longest wait, and Served today
- A visual visit-stage tracker (Checked in → Now serving → Scanning → Complete) showing where the current ticket sits
- “Now serving” live card showing the active ticket number, patient name, scan type, priority, and elapsed time, with Skip and Mark done actions
- A “Waiting – N patients” list showing each queued ticket with its scan type, priority tag, and estimated wait; an URGENT patient is visually flagged in red with a one-tap “Call next” action that jumps them ahead of the queue
- “+ Add walk-in” button in the header to register an unscheduled patient directly into the live queue

Relation to requirements / research proposal

Implements FR3 (“Allow staff to call, skip, or complete tickets and manage the live queue”) and FR4 (“Automatically flag and prioritise urgent/critical patients”) from Table 3.1, and is the staff-side counterpart of the patient-facing Live Queue Tracking screen — an action taken here (e.g. Mark done) is reflected in real time on the patient’s mobile app, per the shared REST API architecture in Figure 3.2. It is the design response to the “no formal queue-numbering system” and “difficulty prioritising urgent cases” problems identified in Chapter 1.

Screen 5. Queue Prediction



Actor: Staff (ECHO unit technicians and shift-planning staff)

Purpose

Surfaces the machine-learning forecasting engine described in the methodology (Section 3.8) directly to staff, turning historical and live queue data into a forward-looking view of expected demand so that staffing and pacing decisions can be made proactively rather than reactively.

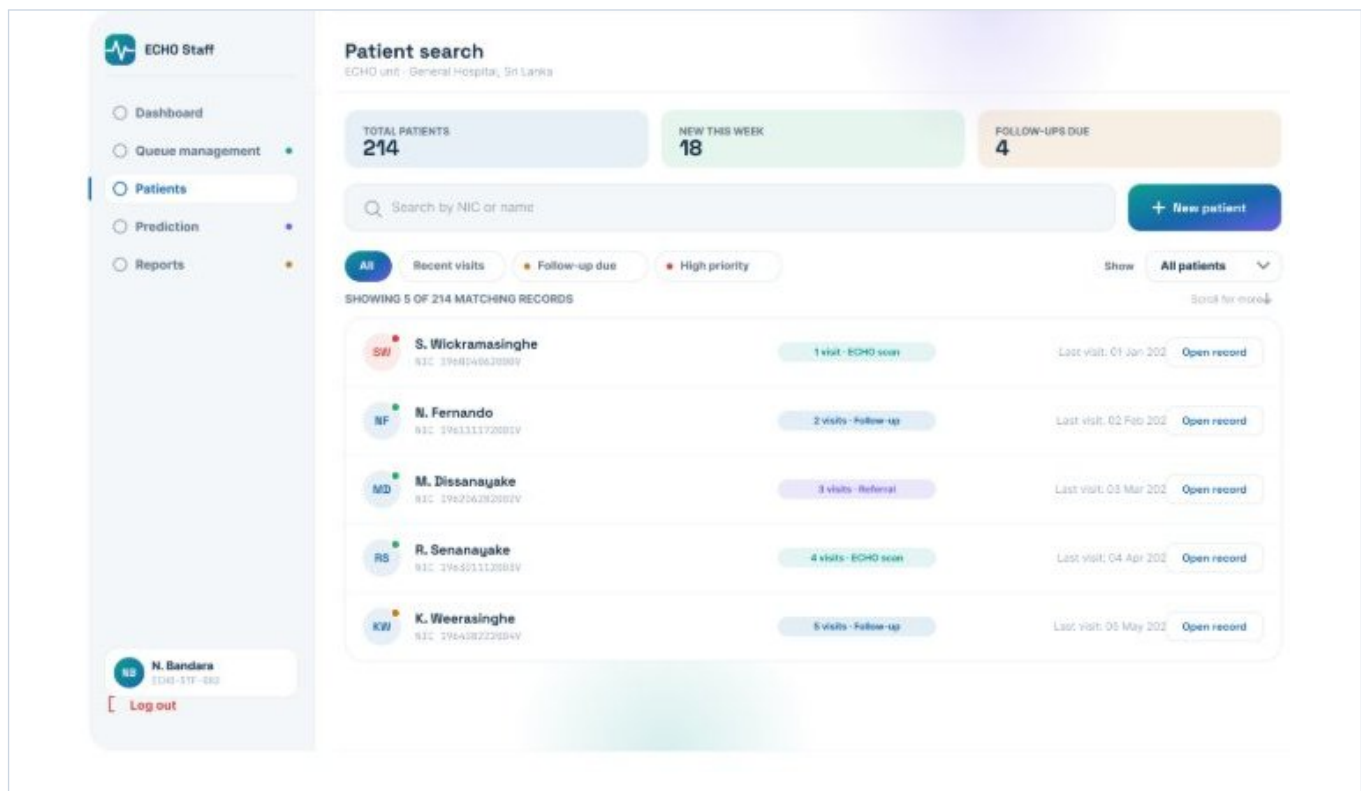
Key functions / elements

- Live model status badge (engine version, last-trained time, live indicator)
- AI Insight banner giving a plain-language recommendation (e.g. inflow trending above average, suggesting an extra technician) with a “View reasoning” link
- KPI cards: patients Expected today, Model confidence, and Next hour forecast
- “Hourly forecast – next 12 hrs” bar chart comparing Actual vs Forecast patient counts per hour, with the busiest/peak hour highlighted
- “What’s driving this forecast” feature-importance panel (day of week, time of day, historical load, local events/holidays) for interpretability
- Model info panel (version, last trained, 7-day accuracy trend, next scheduled retrain) and an automatic retraining-schedule note

Relation to requirements / research proposal

Implements FR5 (“Predict expected patient influx, peak hours, and average wait time using historical/live data”) and is the direct interface for the Random Forest/XGBoost prediction pipeline described in Section 3.8. The feature-importance panel operationalises the interpretability requirement identified from the literature (ref. [14] in Chapter 2), so that staff and administrators can trust and act on the model’s output rather than treating it as a black box.

Screen 6. Patient Search



Actor: Staff (front-desk/registration staff)

Purpose

Digitises patient look-up and registration, replacing the paper-based patient register that is slow to search, prone to loss/damage, and vulnerable to duplicate entries.

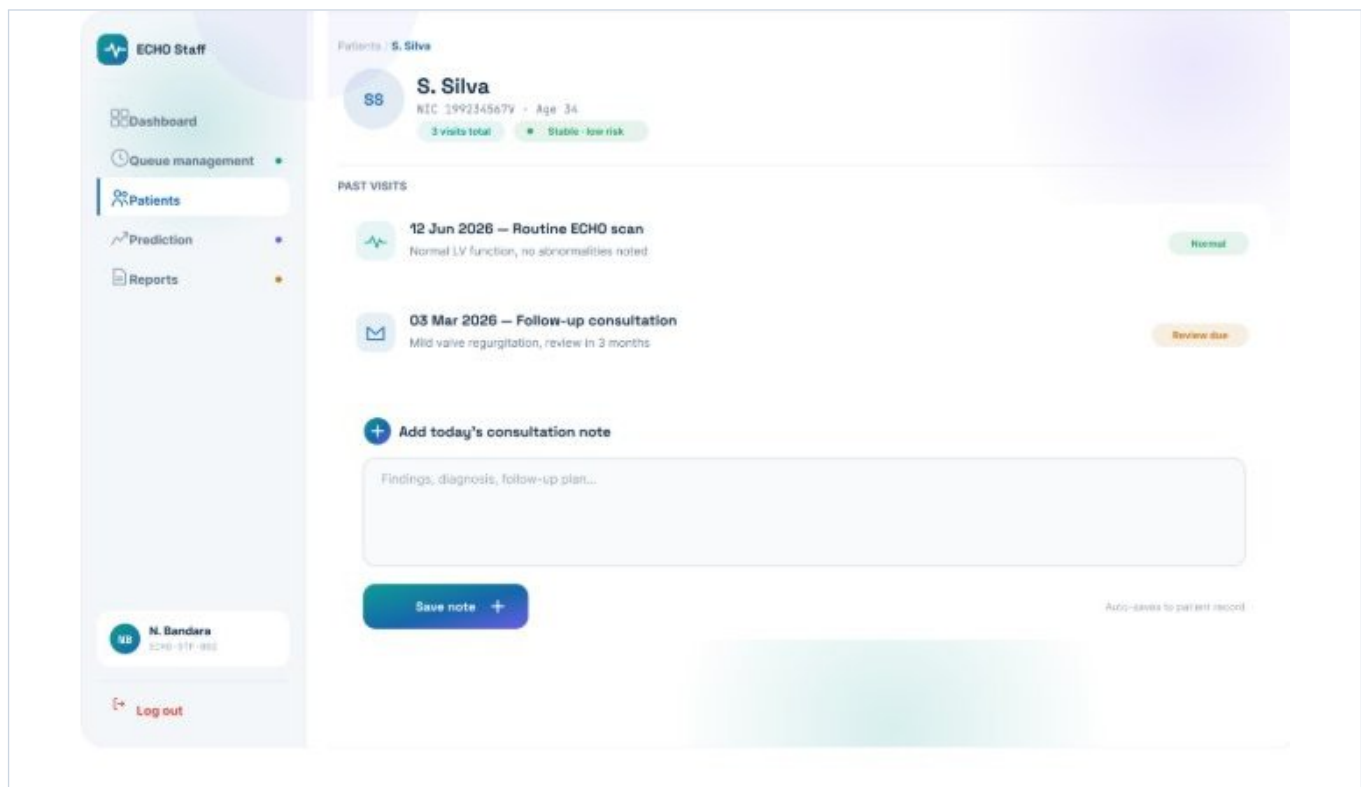
Key functions / elements

- Summary cards: Total patients, New this week, Follow-ups due
- Search bar to find a patient by NIC number or name
- Filter chips: All, Recent visits, Follow-up due, High priority, plus a “Show: All patients” dropdown
- “+ New patient” button that opens the Register New Patient modal (Screen 9)
- A results list showing each patient’s avatar/initials, name, NIC, a colour-coded visit-type tag (e.g. ECHO scan, Follow-up, Referral), last-visit date, and an “Open record” button leading to the Patient Detail screen (Screen 7)

Relation to requirements / research proposal

Implements the search/registration half of FR6 (“Allow staff to search, view, and register patient records, and add new patients to the queue”) and directly targets the “manual patient registration” problem from Chapter 1 (records that are time-consuming to search, prone to loss, and vulnerable to duplication).

Screen 7. Patient Detail



Actor: Staff (medical officers/technicians documenting a visit, and front-desk staff reviewing history)

Purpose

Gives staff a full clinical/visit history for a single patient and a place to record the outcome of today's visit, so patient information persists digitally instead of being scattered across paper files.

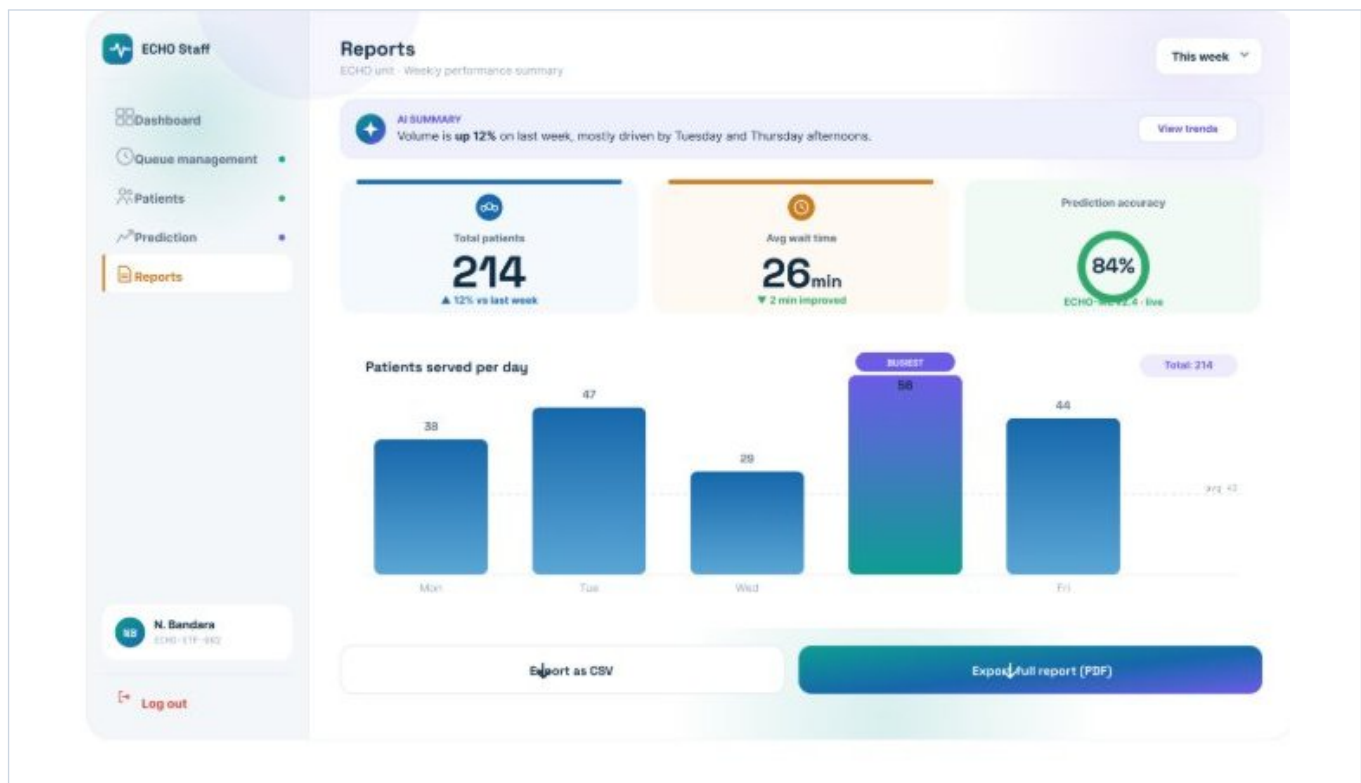
Key functions / elements

- Patient header: avatar/initials, name, NIC, age, total-visits badge, and a status tag (e.g. Stable – low risk)
- “Past visits” timeline listing each prior visit with its date, type, a short clinical summary, and a status badge (e.g. Normal, Review due)
- “Add today’s consultation note” field for findings/diagnosis/follow-up plan, with a “Save note” action that auto-saves to the patient’s record
- Breadcrumb navigation back to the Patient Search list

Relation to requirements / research proposal

Implements the record-viewing half of FR6 and operationalises the shift “from paper-based reporting and absence of historical analysis” toward a searchable digital history, directly supporting continuity of care for repeat ECO patients.

Screen 8. Reports



Actor: Staff (shift leads / ECHO unit technicians compiling operational reports)

Purpose

Replaces manual, paper-compiled monthly/weekly performance reporting with an on-demand, exportable operational reporting screen, enabling evidence-based rather than reactive planning.

Key functions / elements

- Period selector (e.g. “This week”)
- AI Summary banner highlighting a notable trend (e.g. volume up 12% on last week, driven by specific days) with a “View trends” link
- KPI cards: Total patients (with week-on-week change), Avg wait time (with improvement indicator), and live Prediction accuracy
- “Patients served per day” bar chart (Mon–Fri) with the busiest day highlighted and an average reference line
- “Export as CSV” and “Export full report (PDF)” actions

Relation to requirements / research proposal

Implements FR7 (“Generate operational performance reports over a selectable period”) on the staff side (mirrored by the Administrator Portal’s own Reports & Insights screen), and responds to the “paper-based reporting and absence of historical analysis” problem identified in Chapter 1.

Screen 9. Register New Patient (modal)

Actor: Staff (front-desk/registration staff)

Purpose

A focused overlay form for adding a walk-in or newly referred patient straight into today's queue, described in its own subtitle as digitising “the manual sign-in book.”

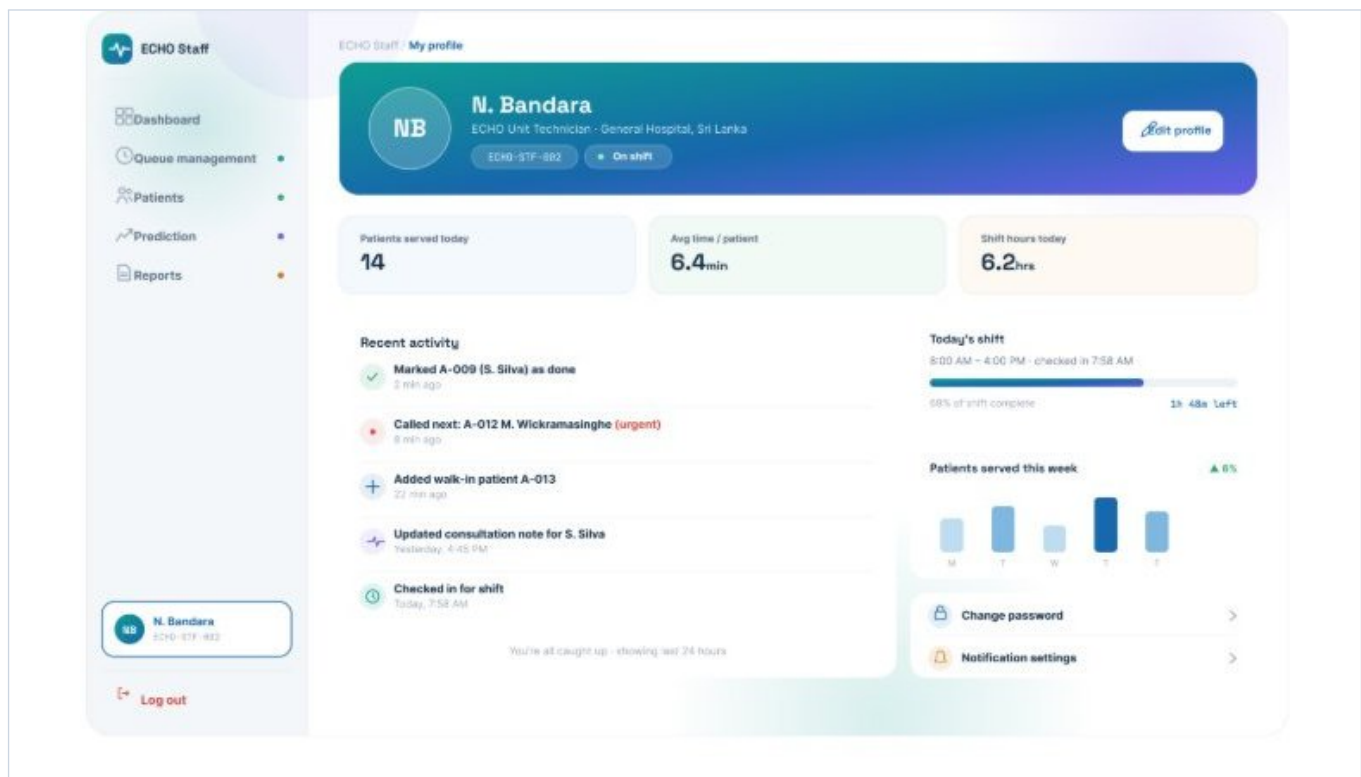
Key functions / elements

- Fields: Full name, NIC number, Phone, Date of birth
- “Reason for visit” dropdown (e.g. Routine ECHO scan)
- Priority selector toggle: Normal or Urgent
- “Cancel” and “Save and add to queue” actions — saving both creates the patient record and issues a live queue ticket in one step

Relation to requirements / research proposal

Implements FR1 (“Allow a patient to digitally join a queue... and receive a ticket”, here performed on the patient's behalf by staff) and the registration half of FR6, and is reachable both from the Patient Search screen (“+ New patient”) and the Queue Management screen (“+ Add walk-in”). It is the clearest single-screen answer to the “manual patient registration” and “no formal queue-numbering system” problems in Chapter 1.

Screen 10. My Profile



Actor: Staff (the individual signed-in staff member, e.g. an ECHO Unit Technician)

Purpose

Gives a staff member visibility into their own shift performance and recent actions within the system, and is the entry point for managing their personal account.

Key functions / elements

- Profile header: avatar/initials, name, role, hospital, Staff ID badge, and on-shift status, with an “Edit profile” button linking to Screen 11
- Personal KPI cards: Patients served today, Avg time/patient, and Shift hours today
- “Recent activity” feed — a chronological, self-scoped log of the staff member’s own actions (e.g. marked a ticket done, called an urgent patient, added a walk-in, updated a consultation note, checked in for shift)
- “Today’s shift” progress bar showing shift start/end time, check-in time, and time remaining
- “Patients served this week” mini bar chart
- “Change password” and “Notification settings” shortcuts

Relation to requirements / research proposal

A supporting account-management screen that complements the Audit Log and role-based access controls discussed in Section 3.10: the personal “Recent activity” feed gives each staff member the same kind of user-attributed action history that, at unit level, feeds into the Administrator Portal’s Audit Log.

Screen 11. Edit Profile

ECHO Staff / My profile · [Edit](#)

Edit profile

Update your personal details and work information

Profile photo
(JPG or PNG, at least 200x200px)

[Remove](#)
[Upload new](#)

Personal information

Full name

Phone

Email address

Date of birth

Work details

Staff ID

Role

Department

Staff ID and role are managed by hospital admin and can't be edited here.

[Cancel](#)
[Save changes](#)

Last updated 3 weeks ago

N. Bandara
ECHO-STF-882

[Log out](#)

Actor: Staff (the individual signed-in staff member)

Purpose

Lets a staff member update their own personal contact details while keeping administratively controlled fields locked, enforcing the role-based governance model described in the methodology.

Key functions / elements

- Profile photo upload with “Remove” and “Upload new” actions
- Editable personal information: Full name, Phone, Email address, Date of birth
- Read-only “Work details” block: Staff ID, Role, and Department, explicitly labelled as “managed by hospital admin and can’t be edited here”
- “Cancel” and “Save changes” actions, with a “Last updated” timestamp

Relation to requirements / research proposal

Reinforces the security/permission model described in Section 3.10: identity- and role-defining fields (Staff ID, Role, Department) can only be changed through the Administrator Web Portal’s Staff Accounts screen, while staff retain self-service control only over their own contact details — a deliberate separation of concerns between individual users and system administrators.